Problem Set 1

Foundations of Machine Learning

- **Submission**: Turn in as a `PDF` and a **separate source code file** on `http://mycourses.unh.edu`. Scanned handwritten notes are encouraged.
- **Questions**: See the class website for a link to office hours

Relevant reading: IRA 1, 2, 6, 8; MML 5.1; CO 1.1, 3.1.1.

1 Recognizing functions

For each of the following functions $f : \mathbb{R} \to \mathbb{R}$, state whether it is *linear*, *affine*, *quadratic*, or none of these. Justify each answer in one sentence.

1. $f(x) = 3x$
2. $f(x) = 3x - 2$
3. $f(x) = x(x - 2)$
4. $f(x) = (x + 1)^2 - x^2$
5. $f(x) = |x|$
6. $f(x) = 2^x$

2 Derivatives warm-up

Compute the first derivative $f'(x)$ and the second derivative $f''(x)$ of each function. Show your work; no credit for unsupported answers.

1. $f(x) = 4x^3 - 2x^2 + 7$
2. $f(x) = e^{-2x}$
3. $f(x) = \log(1 + x^2)$
4. $f(x) = xe^x$
5. $f(x) = \frac{1}{1+e^{-x}}$ (the sigmoid function; express f' in terms of f if you can)
6. $f(x) = (3x - 1)^4$

3 Quadratic functions

Consider the general quadratic function $f(x) = ax^2 + bx + c$ with $a \neq 0$.

1. Derive the location $x^\star$ of the stationary point of f (the point where $f'(x^\star) = 0$).
2. Under what condition on a is $x^\star$ a minimizer? A maximizer?
3. Write $f(x) = 2x^2 - 8x + 3$ in the form $f(x) = a(x - x^\star)^2 + d$ and use this form to read off its minimizer and minimal value.

4 Taylor series by hand

Recall that the first-order and second-order Taylor approximations of f around a point x_0 are

$$f_1(x) = f(x_0) + f'(x_0)(x - x_0), \qquad f_2(x) = f(x_0) + f'(x_0)(x - x_0) + \tfrac{1}{2} f''(x_0)(x - x_0)^2.$$

1. Compute f_1 and f_2 for $f(x) = e^x$ around $x_0 = 0$.
2. Compute f_1 and f_2 for $f(x) = \log(x)$ around $x_0 = 1$.
3. For each function above, evaluate f, f_1, and f_2 at $x = 0.5$ and at $x = 2$, and report the approximation errors $|f(x) - f_1(x)|$ and $|f(x) - f_2(x)|$. Which approximation is better, and where does the approximation degrade?
4. For which functions f does $f_2 = f$ exactly for every choice of x_0? Explain.

5 Plotting functions [code]

1. Plot $f(x) = x^4 - 3x^2 + x$ on the interval $[-2.5, 2.5]$.
2. On the same axes (or a second panel), plot its derivative $f'(x)$. Mark the points where $f'(x) = 0$ and describe, in a sentence or two, how the sign of f' relates to whether f is increasing or decreasing.
3. Plot the sigmoid function $f(x) = 1/(1 + e^{-x})$ on $[-6, 6]$ together with its derivative. Where is the derivative largest?

6 Visualizing Taylor approximations [code]

Let $f(x) = \log(1 + e^x)$.

1. Derive $f'(x)$ and $f''(x)$ by hand and include the derivation in your writeup.
2. Write a function that, given x_0, returns the first-order Taylor approximation f_1 and the second-order Taylor approximation f_2 of f around x_0.
3. Plot f, f_1, and f_2 on $[-4, 4]$ for $x_0 = 0$, and again for $x_0 = 2$, clearly labeling the curves (use a legend).
4. In 2–3 sentences: near x_0, which approximation is more accurate? What happens far from x_0?

7 Finding and classifying critical points

For each function, find all points where $f'(x) = 0$ and classify each as a local minimum, local maximum, or neither, using the second derivative (or another argument when the second derivative is inconclusive).

1. $f(x) = x^2 - 6x + 1$
2. $f(x) = x^3 - 3x$
3. $f(x) = x^3$
4. $f(x) = x^4$
5. $f(x) = x + \frac{4}{x}$ on $x > 0$

8 Convexity

Recall that a twice-differentiable function $f\colon \mathbb{R} \to \mathbb{R}$ is convex if and only if $f''(x) \geq 0$ for all x.

1. Determine which of the following functions are convex on the given domain: (i) $f(x) = e^{3x}$ on $\mathbb{R}$; (ii) $f(x) = \log(x)$ on $x > 0$; (iii) $f(x) = x^4 - x^2$ on $\mathbb{R}$; (iv) $f(x) = \max\{x, 0\}$ on $\mathbb{R}$ (careful: not differentiable at 0; argue from the definition or a picture).
2. Show that if f and g are convex, then $f + g$ is convex, and that αf is convex for any $\alpha \geq 0$.
3. Show that $f(x) = (x-3)^2 + e^x$ is convex and explain why any point with $f'(x) = 0$ must be its global minimizer.
4. Give an example of two convex functions whose *product* is not convex.

9 Time spent

Report approximately how many hours you spent completing this assignment (including the reading).